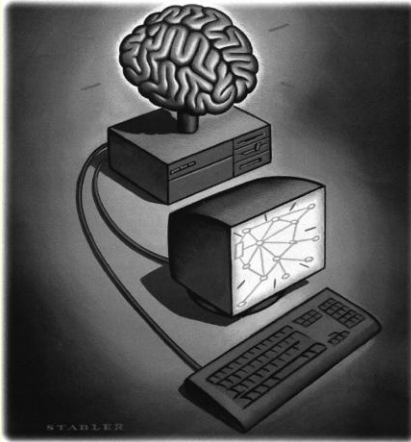




Case Study: Heart Classification

Ali Ridho Barakbah, Entin Martiana

Intelligent Knowledge Creation (iKnow) Research Group
Department of Information and Computer Engineering
Politeknik Elektronika Negeri Surabaya

Heart Dataset

- Source: UCI Repository
- Attributes: 13
 - Age, sex, chest pain type, resting blood pressure, serum cholestoral in mg/dl, fasting blood sugar > 120 mg/dl, resting electrocardiographic results, maximum heart rate achieved, exercise induced angina, oldpeak = ST depression induced by exercise relative to rest, the slope of the peak exercise ST segment, number of major vessels (0-3) colored by flourosopy, thal
- Classes: 2
 - Absence, Presence
- Number of data: 270
 - Absence : 150
 - Presence : 120

Heart Dataset

```
1 70,male,asymptomatic,130,322,0,ventricular_hypertrophy,109,no,2.4,flat,3,normal,Presence
2 67,female,non_anginal_pain,115,564,0,ventricular_hypertrophy,160,no,1.6,flat,0,reversable_defect,Absence
3 57,male,atypical_angina,124,261,0,normal,141,no,0.3,upsloping,0,reversable_defect,Presence
4 64,male,asymptomatic,128,263,0,normal,105,yes,0.2,flat,1,reversable_defect,Absence
5 74,female,atypical_angina,120,269,0,ventricular_hypertrophy,121,yes,0.2,upsloping,1,normal,Absence
6 65,male,asymptomatic,120,177,0,normal,140,no,0.4,upsloping,0,reversable_defect,Absence
7 56,male,non_anginal_pain,130,256,1,ventricular_hypertrophy,142,yes,0.6,flat,1,fixed_defect,Presence
8 59,male,asymptomatic,110,239,0,ventricular_hypertrophy,142,yes,1.2,flat,1,reversable_defect,Presence
9 60,male,asymptomatic,140,293,0,ventricular_hypertrophy,170,no,1.2,flat,2,reversable_defect,Presence
10 63,female,asymptomatic,150,407,0,ventricular_hypertrophy,154,no,4,flat,3,reversable_defect,Presence
11 59,male,asymptomatic,135,234,0,normal,161,no,0.5,flat,0,reversable_defect,Absence
12 53,male,asymptomatic,142,226,0,ventricular_hypertrophy,111,yes,0,upsloping,0,reversable_defect,Absence
13 44,male,non_anginal_pain,140,235,0,ventricular_hypertrophy,180,no,0,upsloping,0,normal,Absence
14 61,male,typical_angina,134,234,0,normal,145,no,2.6,flat,2,normal,Presence
15 57,female,asymptomatic,128,303,0,ventricular_hypertrophy,159,no,0,upsloping,1,normal,Absence
16 71,female,asymptomatic,112,149,0,normal,125,no,1.6,flat,0,normal,Absence
17 46,male,asymptomatic,140,311,0,normal,120,yes,1.8,flat,2,reversable_defect,Presence
18 53,male,asymptomatic,140,203,1,ventricular_hypertrophy,155,yes,3.1,downsloping,0,reversable_defect,Presence
19 64,male,typical_angina,110,211,0,ventricular_hypertrophy,144,yes,1.8,flat,0,normal,Absence
20 40,male,typical_angina,140,199,0,normal,178,yes,1.4,upsloping,0,reversable_defect,Absence
21 67,male,asymptomatic,120,229,0,ventricular_hypertrophy,129,yes,2.6,flat,2,reversable_defect,Presence
22 48,male,atypical_angina,130,245,0,ventricular_hypertrophy,180,no,0.2,flat,0,normal,Absence
23 43,male,asymptomatic,115,303,0,normal,181,no,1.2,flat,0,normal,Absence
24 47,male,asymptomatic,112,204,0,normal,143,no,0.1,upsloping,0,normal,Absence
25 54,female,atypical_angina,132,288,1,ventricular_hypertrophy,159,yes,0,upsloping,1,normal,Absence
26 48,female,non_anginal_pain,130,275,0,normal,139,no,0.2,upsloping,0,normal,Absence
27 46,female,asymptomatic,138,243,0,ventricular_hypertrophy,152,yes,0,flat,0,normal,Absence
28 51,female,non_anginal_pain,120,295,0,ventricular_hypertrophy,157,no,0.6,upsloping,0,normal,Absence
29 58,male,non_anginal_pain,112,230,0,ventricular_hypertrophy,165,no,2.5,flat,1,reversable_defect,Presence
30 71,female,non_anginal_pain,110,265,1,ventricular_hypertrophy,130,no,0,upsloping,1,normal,Absence
31 57,male,non_anginal_pain,128,229,0,ventricular_hypertrophy,150,no,0.4,flat,1,reversable_defect,Presence
32 66,male,asymptomatic,160,228,0,ventricular_hypertrophy,138,no,2.3,upsloping,0,fixed_defect,Absence
33 37,female,non_anginal_pain,120,215,0,normal,170,no,0,upsloping,0,normal,Absence
```

Latihan 7 - #Assignment

- Bacalah dataset heart.csv dan konversikan data kategori menjadi data numerik
- Lakukan normalisasi dengan Min-Max (0-1), Z-Score dan Sigmoidal
- Lakukan klasifikasi terhadap dataset dengan k-NN ($k=1$, $k=5$, $k=9$) dan Decision Tree
- Pakailah validation model k-Fold ($k=4$, $k=10$) dan LOO
- Hitunglah error ratio dari setiap percobaan diatas

Pengumpulan Tugas

- Buatlah coding dengan Bahasa pemrograman/tools apapun untuk semua assignment
- Buatlah laporan dalam slide ppt. Laporan terdiri dari screenshot coding dan hasil running untuk setiap assignment.
- Simpan laporan dalam file pdf dengan format penamaan:
DM_M7_NRP_namadepan.pdf
- Deadline upload:
 - Reguler & RPL Kelas A : Selasa, 6 Oktober 2026 pk 23.00
 - RPL Kelas B : Sabtu, 10 Oktober 2026 pk 23.00